

Calibration of Shortfall, Wastage, and Transmission Operating Costs in the Nested WECC–California Model

ASTR2026 transmission layer

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1 Purpose

The nested WECC–California linear program prices three quantities that are not physical generation: *shortfall* (unserved demand), *wastage* (curtailed surplus), and *transmission operating cost* (per-corridor flow cost). These are modelling parameters rather than measured data, and their values determine how the optimiser trades unserved energy, curtailment, and wheeling against real generation cost.

This document records the benchmarks used to set them, the derivations, and the diagnosis that motivated the recalibration.

2 Unit convention

All costs in the model are expressed in dollars per joule, \$/J. Published benchmarks are almost universally quoted in dollars per megawatt hour. The conversion is

$$1 \text{ MWh} = 10^6 \text{ W} \times 3600 \text{ s} = 3.6 \times 10^9 \text{ J}, \quad \left[\frac{\$}{\text{MWh}} \right] = \left[\frac{\$}{\text{J}} \right] \times 3.6 \times 10^9. \quad (1)$$

Failing to apply this factor is what produced the miscalibration described in Section 3.

3 Diagnosis: the miscalibration

The generator operating costs carried in the WECC graph are correctly scaled: nuclear at 3.09×10^{-9} \$/J is \$11.13/MWh, and the most expensive unit in the fleet at 1.07×10^{-8} \$/J is \$38.52/MWh. Both are realistic. The penalty parameters were not on the same scale.

Table 1: Pre-calibration parameter values expressed in \$/MWh.

Parameter	\$/J (previous)	= \$/MWh	Benchmark
<code>shortfall_cost</code>	1×10^{-2}	36,000,000	\$10,000/MWh
<code>wastage_cost</code>	1×10^{-4}	360,000	\$0–25/MWh
Transmission (meso)	0	0	> 0 required

Two consequences were measured on the 672-hour S0 solve.

The objective was almost entirely penalty. With shortfall of 36.523 GWh and wastage of 592.954 GWh:

$$\text{shortfall penalty} = 36.523 \text{ GWh} \times \$3.6 \times 10^7/\text{MWh} = \$1.315 \times 10^{12}, \quad (2)$$

$$\text{wastage penalty} = 592.954 \text{ GWh} \times \$3.6 \times 10^5/\text{MWh} = \$0.213 \times 10^{12}, \quad (3)$$

$$\text{sum} = \$1.528 \times 10^{12}, \quad (4)$$

against a total objective of $\$1.528569 \times 10^{12}$. More than 99.9% of the objective was these two penalties. Real generation cost — which determines which plants run, and therefore CO₂ — accounted for roughly 0.04%.

The LP was badly conditioned. Objective coefficients spanned 1.98×10^{-13} to 1×10^{-2} , about eleven orders of magnitude. Gurobi reported coefficient-range warnings on every solve, and the S3 scenario terminated with **Numerical trouble encountered**.

Because generation cost was numerically negligible, the dispatch decision was close to degenerate: the solver had little incentive to distinguish between generators. This is consistent with the observed sensitivity of per-asset results to the Gurobi **Crossover** setting, in which barrier-only and crossover solutions agreed on the objective to 0.04% but differed by a factor of 2.6 in CO₂ and 3.5 in fossil generation.

4 Calibrated values

Table 2: Calibrated parameters, benchmarks, and sources.

Parameter	Value (\$/J)	= \$/MWh	Benchmark and rationale	Source
<code>shortfall_cost</code> (VOLL)	2.78×10^{-6}	10,000	Value of lost load. \$10,000/MWh is the default VOLL in the GenX capacity expansion model and sits in the lower-middle of the \$9,000–\$45,000/MWh range reported for developed economies. ERCOT adopted \$35,000/MWh; EU member state values range from EUR 4,000 to EUR 69,000/MWh. Retained at \$10,000/MWh as a conservative, widely used default; still about 260× the priciest generator, so load is served whenever physically possible.	[1, 2, 3, 4, 5]

Parameter	Value (\$/J)	= \$/MWh	Benchmark and rationale	Source
wastage_cost (curtailment)	2.78×10^{-10}	1.00	Curtailment of surplus is conventionally valued at \$0/MWh. A strictly positive value is nonetheless required here as a tie-breaker: at exactly zero, corridor arc pairs form a flat direction in the objective that an interior-point method smears across. \$1/MWh is the low end of the curtailment range and far below any generator's cost, so it breaks ties without steering dispatch.	[6, 7]
Transmission operating cost (meso corridors)	1.0×10^{-10}	0.36	Published wheeling charges are about \$2/MWh, but price a whole point-to-point transaction rather than each line crossed. The BA tier charges transaction-scale rates (\$8/MWh on 40 of 68 lines) because BA paths are few hops; meso paths cross many of about 5,200 substation corridors, so a transaction-scale per-hop rate would dominate merit order. \$0.36/MWh per corridor puts a ten-hop intra-CA path near \$3.6/MWh — the right order of magnitude for a transaction, and well below gas (about \$38/MWh).	[8, 9]

4.1 Why transmission cost must be strictly positive

Each corridor is represented as two opposing directed arcs, because the **Transmission** class holds a single non-negative flow variable and applies its efficiency only on the receiving side. No linear constraint can forbid both arcs of a pair from being active simultaneously. With a zero flow cost, a corridor pair is therefore a perfectly flat direction in the objective: energy may be cycled $A \rightarrow B \rightarrow A$ indefinitely, burning $(1 - \eta)$ per pass at no cost.

A strictly positive per-unit flow cost removes the flatness. Cancelling the common component δ of any bidirectional pair strictly lowers cost by $2\delta c$ and returns the round-trip losses $(1 - \eta)\delta$ to both end nodes, so simultaneous bidirectional flow can never be optimal. The base WECC graph achieves the same effect by being lossless ($\eta = 1$) while charging 2.222×10^{-9} \$/J on most

BA lines.

Introducing the cost reduced gross transmission flow in the crossover solution from 293.3 TWh to 28.7 TWh, and corridors carrying flow in both directions simultaneously from 1,097 to 268, while changing CO₂ by 0.1%.

5 Expected effects of recalibration

1. The objective coefficient range narrows from about eleven to about seven orders of magnitude, improving conditioning.
2. Generation cost becomes the dominant term, so dispatch is driven by generator economics rather than by penalty parameters. This is expected to reduce, though not necessarily eliminate, the degeneracy underlying the **Crossover** sensitivity.
3. Objective values become interpretable in dollars.
4. Shortfall at \$10,000/MWh remains about 260× the most expensive generator, so unserved energy appears only where demand is genuinely infeasible to serve, not as an economic substitute for generation.

6 Recommended sensitivity runs

- **VOLL:** \$35,000/MWh (9.72×10^{-6} \$/J), the ERCOT value, to test whether conclusions depend on the penalty for unserved energy.
- **Curtailement:** \$25/MWh (6.94×10^{-9} \$/J), the wind production tax credit opportunity cost, which is the economically grounded alternative to a pure tie-breaker.
- **Transmission:** \$1/MWh per corridor (2.78×10^{-10} \$/J) to test sensitivity of flow patterns to the wheeling assumption.

7 Where these are set

```
ev_charging_project/config.py
    NETWORK_KW['shortfall_cost'] = 10_000 / 3.6e9 # 2.78e-6 $/J
    NETWORK_KW['wastage_cost'] = 1 / 3.6e9 # 2.78e-10 $/J

Distribution-Grid-EV-CA/09_02_nest_meso_in_good.py
    TRANSMISSION_OPERATING_COST = 0.36 / 3.6e9 # 1.0e-10 $/J
```

10_01_run_scenarios_S0_S3.py no longer overrides the shortfall and wastage costs; **config.py** is the single source of truth.

References

- [1] GenX Project. *GenX: a configurable power system capacity expansion model for studying low-carbon energy futures*. MIT Energy Initiative. <https://github.com/GenXProject/GenX.jl>; <https://genx.mit.edu>. Default value of lost load \$10,000/MWh.
- [2] The Brattle Group. *Value of Lost Load Study for the ERCOT Region*, Project No. 55837, 2024. <https://www.brattle.com/wp-content/uploads/2024/09/Value-of-Lost-Load-Study-for-the-ERCOT-Region.pdf>

- [3] ERCOT. *Estimating the Value of Lost Load: Literature Review and Macroeconomic Analysis*, 2013. https://www.ercot.com/files/docs/2013/06/19/ercot_valueoflostload_literaturereviewandmacroeconomic.pdf
- [4] MISO. *Value of Lost Load (VOLL) Development and Use*, RECBWG, August 2022. <https://cdn.misoenergy.org/>
- [5] Emissions-EUETS. *Value of Lost Load (VoLL)*, internal electricity market glossary. <https://www.emissions-euets.com/internal-electricity-market-glossary/966-value-of-lost-load-voll> EU member state VOLL values.
- [6] Flex Power. *The Economics of Curtailing Renewables like Wind and Solar*. <https://flex-power.energy/energyblog/the-economics-of-curtailing-renewables/> Zero-cost curtailment convention and PTC-driven negative bidding.
- [7] Resources for the Future. *What Are the Costs and Values of Wind and Solar Power? How Are They Changing?* <https://www.resources.org/common-resources/what-are-costs-and-values-wind-and-solar-power-how-are-they-changing/> Production tax credit of about \$25/MWh as curtailment opportunity cost.
- [8] Brown, P. et al. *Optimal transmission expansion modestly reduces decarbonization costs of U.S. electricity*, arXiv:2402.14189. <https://arxiv.org/pdf/2402.14189> Uses a \$2.00/MWh wheeling rate.
- [9] Lawrence Berkeley National Laboratory. *Improving estimates of transmission capital costs for utility-scale wind and solar projects to inform renewable energy policy*, 2019. https://eta-publications.lbl.gov/sites/default/files/td_costs_formatted_final.pdf